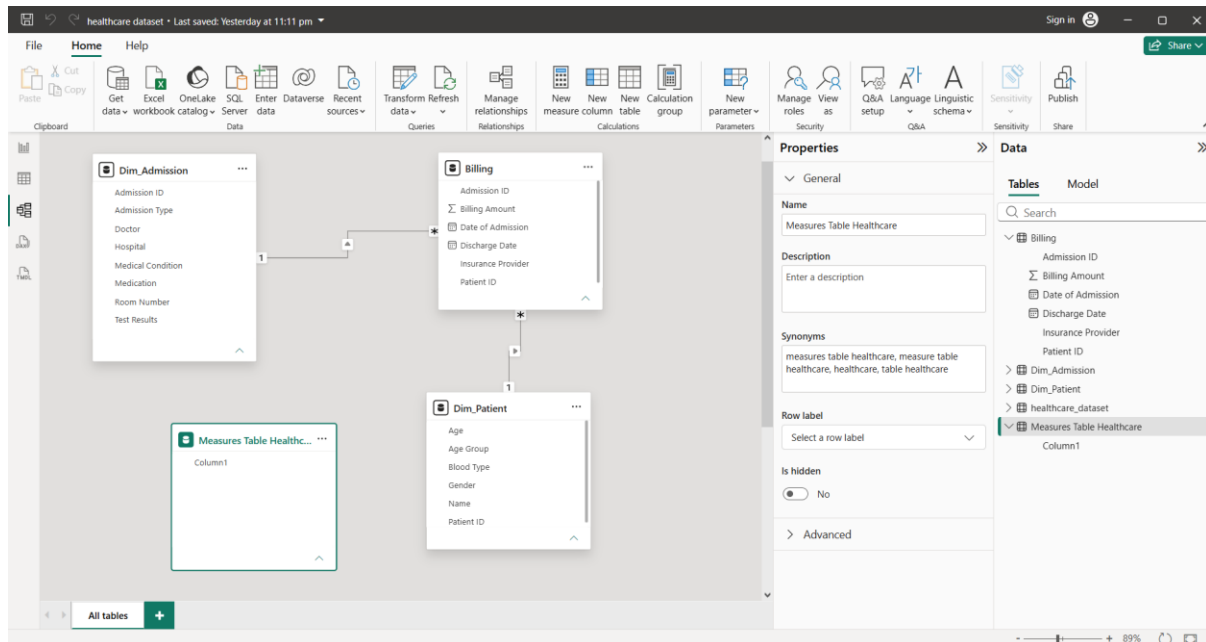
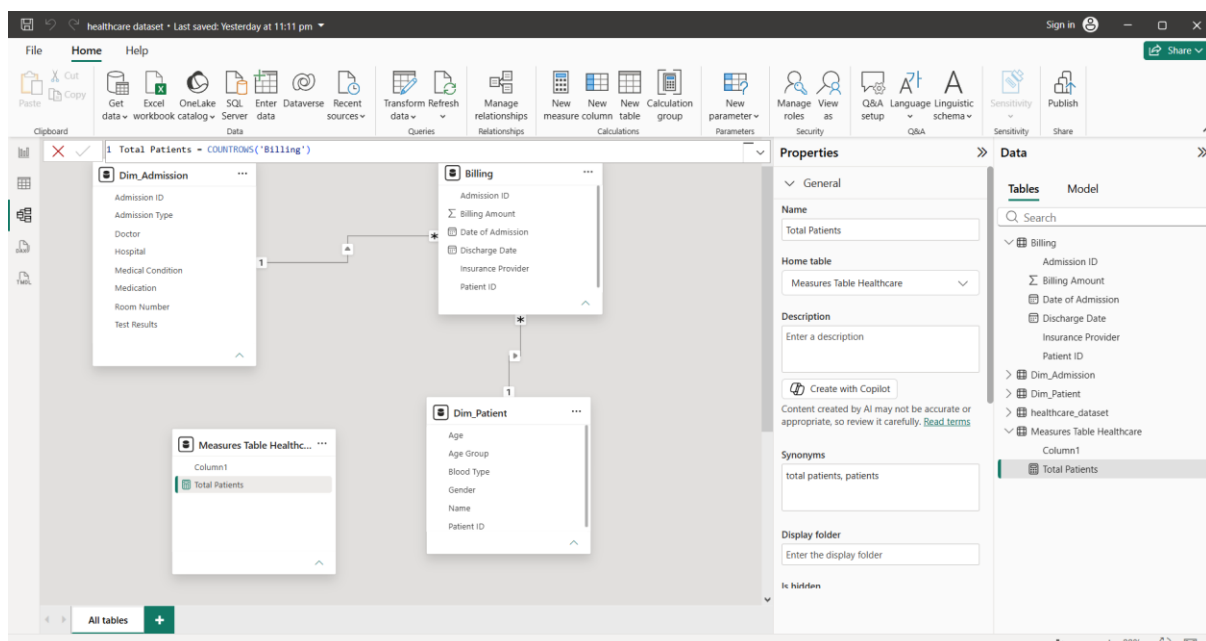


Data Model and Measures Table



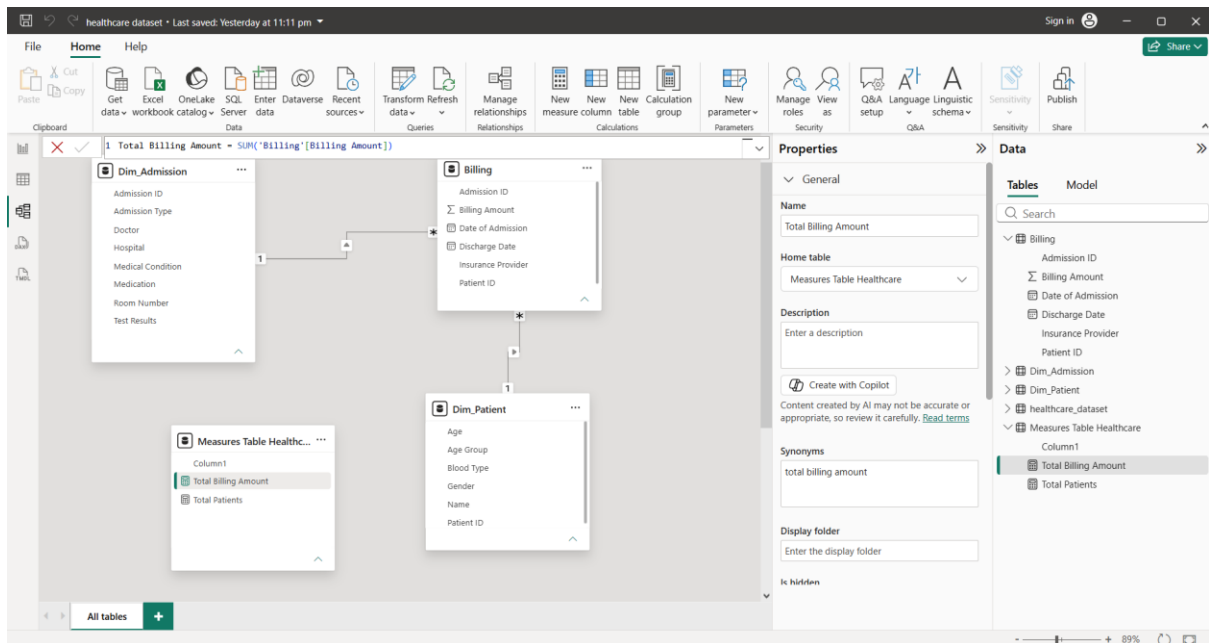
The prepared Week 5 data model was used as the base for Week 6. A separate Measures Table Healthcare was created to store the DAX measures used for the dashboard. The model contains Dim_Patient, Dim_Admission and Billing tables.

Total Patients



The Total Patients measure was created using COUNTROWS on the Billing table. This measure is used to calculate the total number of records available in the billing data.

Total Billing Amount



The Total Billing Amount measure was created using the SUM function on the Billing Amount column. It calculates the total billing amount and is used in the dashboard for overall billing analysis.

Average Billing Amount

The screenshot displays the Microsoft Power BI Desktop interface. The main workspace shows a data model with three tables: **Dim_Admission**, **Billing**, and **Dim_Patient**. **Dim_Admission** is connected to **Billing** via a one-to-many relationship on the **Admission ID** field. **Billing** is connected to **Dim_Patient** via a one-to-many relationship on the **Patient ID** field. A measure named **Avg Billing Amount** is defined in the **Measures Table Healthcare** table, with the formula: `AVERAGE('Billing'[Billing Amount])`. The **Properties** pane on the right shows the measure's configuration: Name is **Avg Billing Amount**, Home table is **Measures Table Healthcare**, and the description is **Enter a description**. The **Data** pane on the right shows the measure is available in the **Measures Table Healthcare** table.

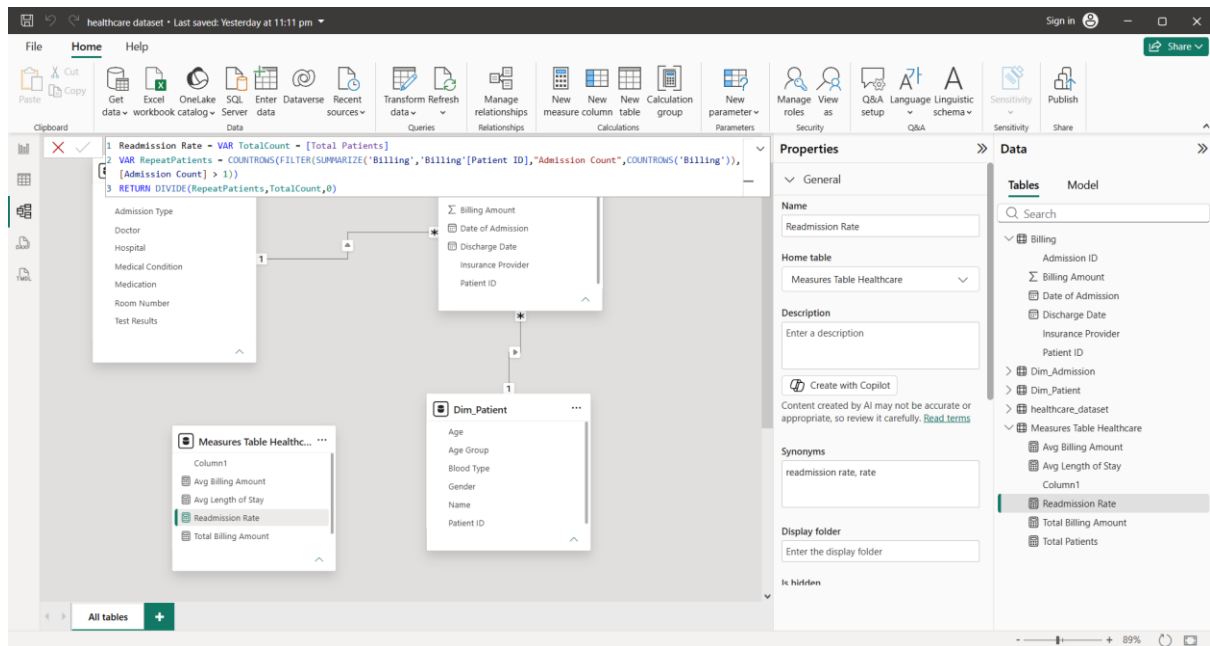
The Avg Billing Amount measure was created using the AVERAGE function on the Billing Amount column. It provides the average billing amount across the available billing records.

Average Length of Stay

The screenshot displays the Microsoft Power BI Desktop interface. The main workspace shows the same data model as the previous screenshot. A measure named **Avg Length of Stay** is defined in the **Measures Table Healthcare** table, with the formula: `AVERAGE('Billing',DATEDIFF('Billing'[Date of Admission],'Billing'[Discharge Date],DAY))`. The **Properties** pane on the right shows the measure's configuration: Name is **Avg Length of Stay**, Home table is **Measures Table Healthcare**, and the description is **Enter a description**. The **Data** pane on the right shows the measure is available in the **Measures Table Healthcare** table.

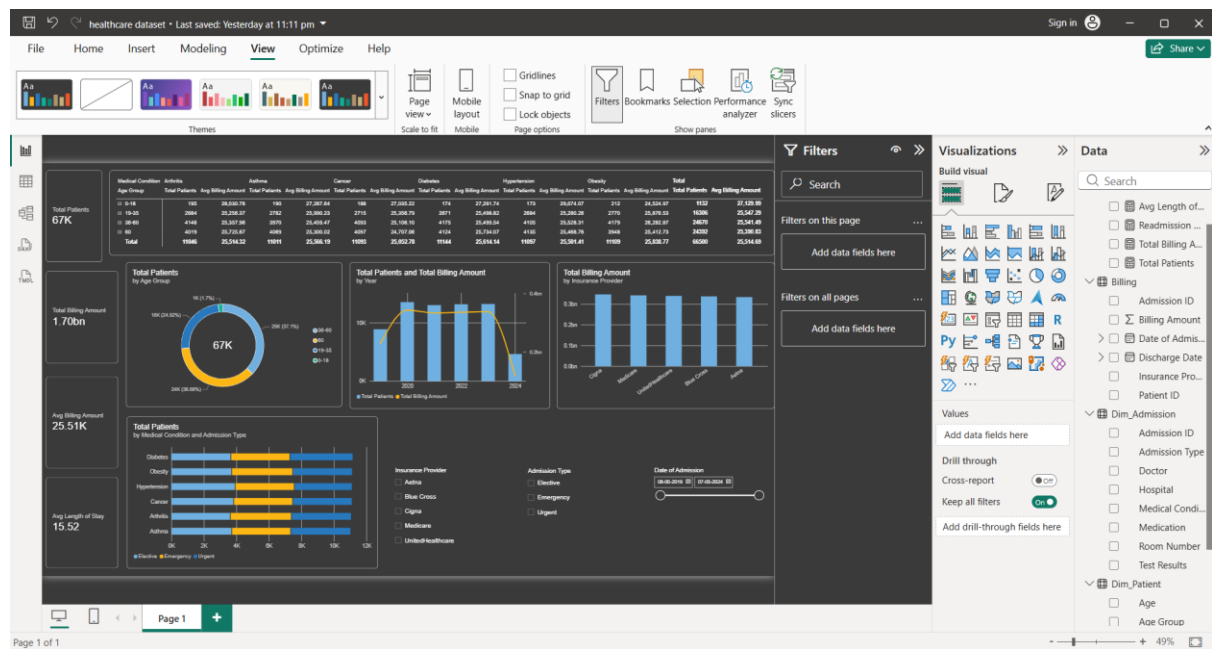
The Avg Length of Stay measure was created using AVERAGEX and DATEDIFF. It calculates the number of days between the Date of Admission and Discharge Date and then finds the average length of stay.

Readmission Rate



A Readmission Rate measure was also created to identify patients with more than one admission. The measure compares repeat patients with the total patient count to calculate the readmission rate.

Final Healthcare Dashboard



The final dashboard combines the created DAX measures with different Power BI visuals.

The dashboard includes:

- Cards for Total Patients, Total Billing Amount, Average Billing Amount and Average Length of Stay
- Matrix showing patients by Age Group, Gender and Medical Condition
- Donut Chart showing patients by Age Group
- Stacked Bar Chart showing Medical Condition by Admission Type
- Line and Stacked Column Chart showing Total Patients and Total Billing Amount over time
- Clustered Column Chart showing Total Billing Amount by Insurance Provider
- Slicers for Insurance Provider, Admission Type and Date of Admission

The slicers make the dashboard interactive and allow the displayed information to be filtered based on the selected criteria.